

EXPERIMENT 33

Denial-of-Service Pattern Analysis

AIM

To identify concentrated high-frequency traffic toward a single host without claiming that the pattern alone proves an attack.

ALGORITHM

1. Create packet-level records.
2. Count packets by destination.
3. Calculate the frequency for each destination.
4. Compare counts with a selected threshold.
5. Report concentrated traffic patterns as possible indicators.

CODE

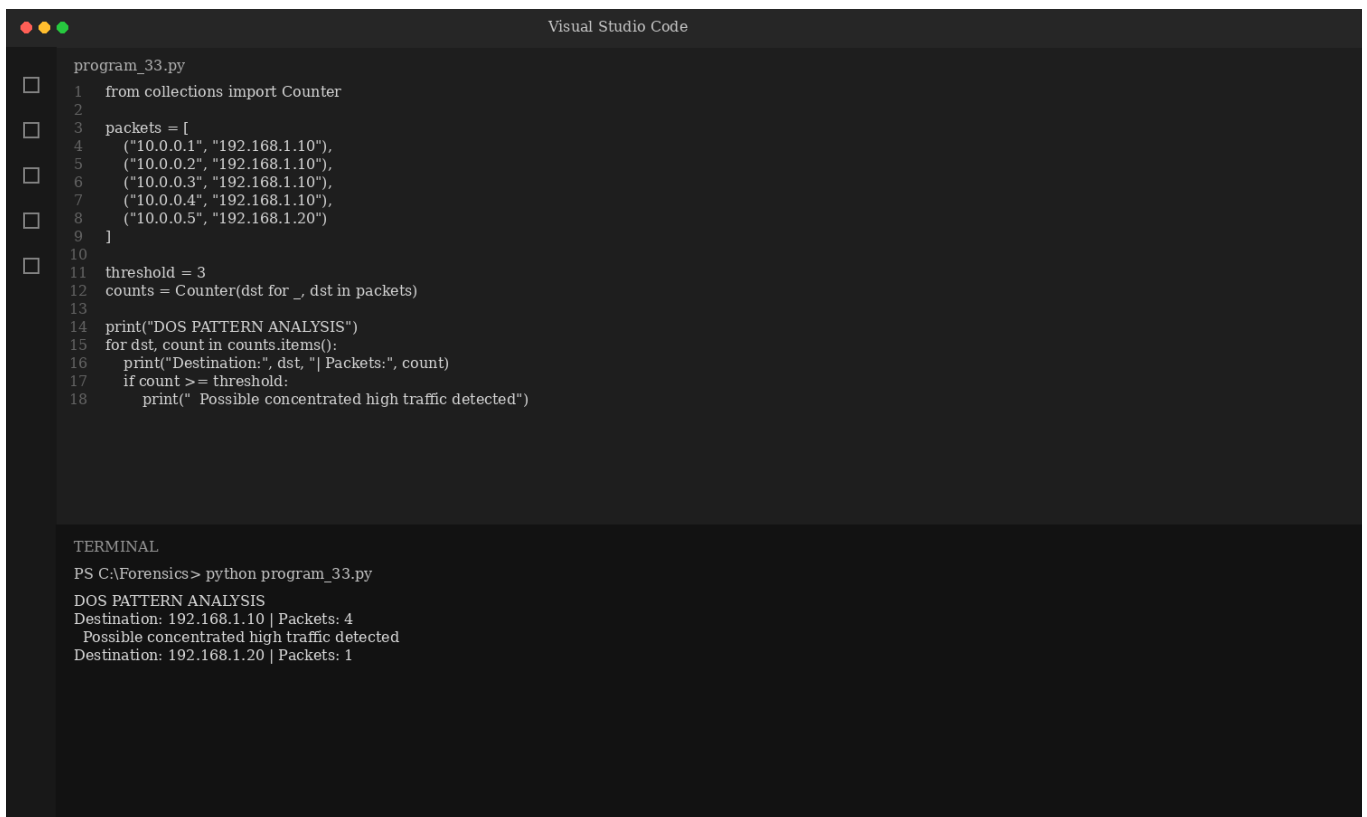
```
from collections import Counter

packets = [
    ("10.0.0.1", "192.168.1.10"),
    ("10.0.0.2", "192.168.1.10"),
    ("10.0.0.3", "192.168.1.10"),
    ("10.0.0.4", "192.168.1.10"),
    ("10.0.0.5", "192.168.1.20")
]

threshold = 3
counts = Counter(dst for _, dst in packets)

print("DOS PATTERN ANALYSIS")
for dst, count in counts.items():
    print("Destination:", dst, "| Packets:", count)
    if count >= threshold:
        print(" Possible concentrated high traffic detected")
```

OUTPUT



The screenshot shows a Visual Studio Code window with a file named 'program_33.py'. The code in the editor is as follows:

```
1 from collections import Counter
2
3 packets = [
4     ("10.0.0.1", "192.168.1.10"),
5     ("10.0.0.2", "192.168.1.10"),
6     ("10.0.0.3", "192.168.1.10"),
7     ("10.0.0.4", "192.168.1.10"),
8     ("10.0.0.5", "192.168.1.20")
9 ]
10
11 threshold = 3
12 counts = Counter(dst for _, dst in packets)
13
14 print("DOS PATTERN ANALYSIS")
15 for dst, count in counts.items():
16     print("Destination:", dst, "| Packets:", count)
17     if count >= threshold:
18         print(" Possible concentrated high traffic detected")
```

Below the code editor is a terminal window with the following output:

```
PS C:\Forensics> python program_33.py
DOS PATTERN ANALYSIS
Destination: 192.168.1.10 | Packets: 4
 Possible concentrated high traffic detected
Destination: 192.168.1.20 | Packets: 1
```

RESULT

The program identifies concentrated high-frequency traffic that may indicate a denial-of-service pattern.